ROM-fed Fault-Injection Shell — Vivado Elaborated Schematics

Local demonstration / GitHub schematic companion | Vivado 2022.1 | part **xc7vx690tffg1761-2**
Top: `top_fi_rom_*` (example: `top_fi_rom_s3` → DUT `top_s3_subfft_ecc`)

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1. Purpose

These figures are Vivado **Elaborated Design** schematics of the **ROM-fed FI shell** used for local board-style demonstration. On-chip stimulus ROM provides FFT input beats; UART is used **only** for fault-injection (ARM/FIRE) commands. This shell is **not** a paper fair-kernel resource top (`vivado/K_*` / Yosys tables remain authoritative). Do not treat pin counts or utilization from this shell as manuscript numbers.

2. Architecture reading of the two schematics

Figure 1 (shell top) shows the intended separation of paths:

- **Control / FI path:** `uart_rx` → `u_uart` (`uart_rx_simple`) → `u_dec` (`uart_cmd_decoder`) → `u_fi` (`fi_controller`) → DUT FI ports (`fi_fire_pulse`, `fi_arm_*`, `fi_site_mem`).
- **Data / stimulus path:** `u_rom` (`stim_rom`, init from `.mem/.coe`) ← addressed by `u_feed` (`stim_feeder`) → DUT parallel inputs (`in_valid`, `in_last`, lane `in*_re/im`).
- **External ports** stay small: `clk`, `rst`, `uart_rx`, plus status / `out_valid` / `out_last`. Wide FFT I/O remains *inside* the chip.

Figure 2 (DUT interior) is the opened hierarchy under `u_dut`: a left-to-right chain of per-stage blocks (ten radix-2 SDF stages for the 1024-point kernel). Clock/reset fan out to each stage; data and valid handshake propagate stage by stage. This view is the structural counterpart of the per-stage recovery narrative; FI hooks target declared stage/site boundaries rather than arbitrary I/O pins.

Note: Screenshot resolution follows the Vivado schematic canvas. For repository packaging, place this PDF under the public `schematic/` folder alongside other architecture figures.

Figure 1 — ROM-fed FI shell (elaborated top)

Instances: `u_uart`, `u_dec`, `u_fi`, `u_rom`, `u_feed`, `u_dut`.

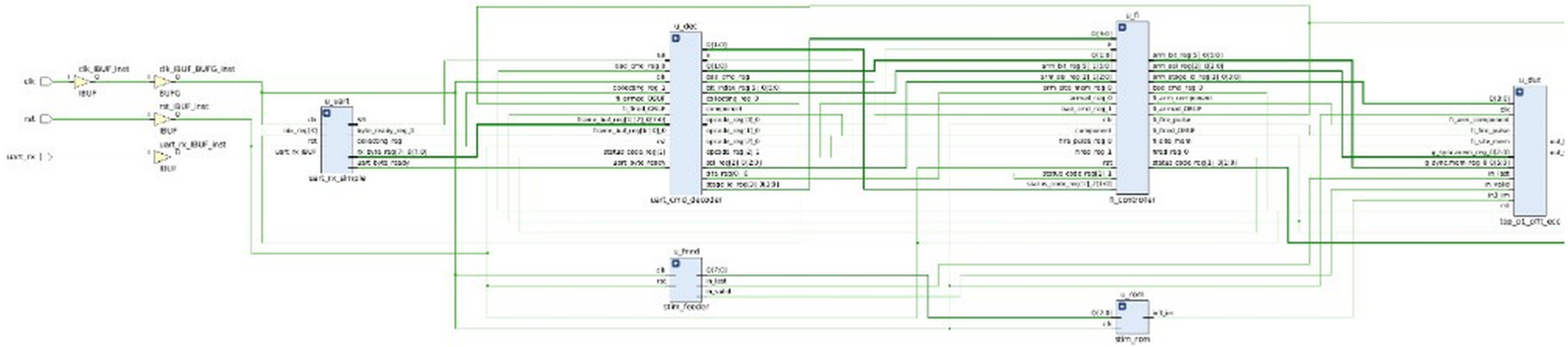

Fig. 1. Vivado elaborated schematic of the ROM-fed FI shell. UART path is FI-only; stimulus ROM feeds the DUT data path.

Figure 2 — DUT stage pipeline (hierarchy under u_dut)

Per-stage SDF chain inside the kernel top (example DUT: `top_s3_subfft_ecc`).

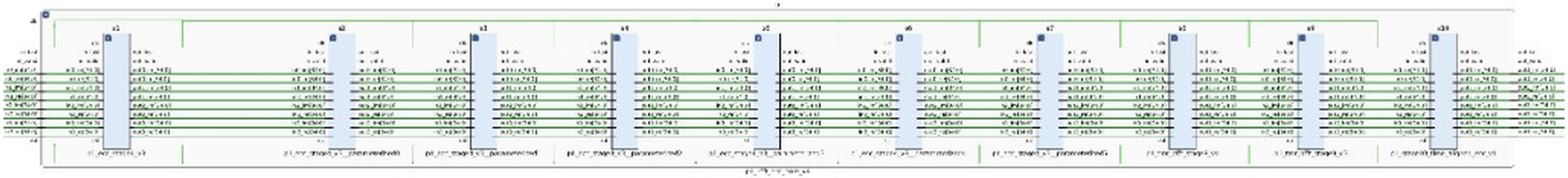

Fig. 2. Vivado schematic of the DUT internal stage pipeline (stages cascaded left → right).

Provenance. Captured from Vivado 2022.1 Elaborated Design views of `fi_rom_top_v1/ROM_*` projects. Stimulus init files: `fi_func_sim_v1/common/coe|mem/*_input.*`. Manuscript fair-kernel RTL remains under `artifact_v1/*_src` / `vivado/K_*`.